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A pan-latitudinal Rodinia in the Tonian true polar wander frame

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ABSTRACT

A new paleomagnetic and geochronological study was conducted on the Tonian Chéngjiāng Formation from Yúnnán Province, South China, to examine the recent reported true polar wander event (TPW) during 825–750 Ma and clarify the position of South China in the Rodinia supercontinent. Detailed thermal demagnetization revealed two remanent magnetic components. A low-temperature component separated below 300 °C is interpreted as a recent viscous remanence. Additionally, a high-temperature component (H1) with unblocking temperature up to 690 °C is revealed. Passing both the fold test and reversal test, and recording at least two and half magnetochrons, H1 is suggested to be a primary remanence. The paleomagnetic pole (CJH1, 33.4°N, 56.1°E, $dp/dm = 7.3°/8.9°$) calculated from component H1 has no similarities to the Phanerozoic apparent polar wander path of South China. SHRIMP II U-Pb zircon data from a tuff bed near the paleomagnetic sampling sites suggest that the H1 pole has an age of ca. 800 Ma. Combining the reported Tonian paleomagnetic results from South China and Svalbard, we suggest a ~63° TPW event during 825–790 Ma. Under this TPW paleomagnetic frame and using the reliable Tonian paleomagnetic results from other continents, we reconstruct the main part of Rodinia. This new configuration of Rodinia reconstructs South China and India at its northern periphery, and within the polar or high latitude zone during ca. 900–750 Ma. The polar location of those cratons, together with the mid-high latitudinal distribution of Australia, Congo and Baltica, suggests that Rodinia was a pan-latitudinal rather than an equatorially distributed supercontinent. The reconstruction also suggests that the breakup of the Rodinia should have been after 750 Ma but before 720 Ma. Despite similarities of their assembly and configuration, Rodinia and Pangea affected the Earth system differently.
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1. Introduction

The Neoproterozoic supercontinent Rodinia likely had critical impacts on the evolution of Earth system, e.g. animal evolution, extreme climate change, and tectonic evolution (Hoffman et al., 2017, 1998). Although intense research from many disciplines has been conducted through almost three decades, Rodinia's duration and configuration are still debated (Evans, 2009; Jiang et al., 2003; Karlstrom et al., 1999; Li et al., 1995, 2004, 2008; Merdith et al., 2017b, 2017a; Moores, 1991; Wen et al., 2018; Wingate et al., 2002). Recently, some researchers (Li et al., 2004; Maloof et al., 2006; Niu et al., 2016; Swanson-Hysell et al., 2012) hypothesized a true polar wander event (TPW) at ca. 800 Ma, which may be used to refine the configuration of Rodinia more accurately and help understanding long-term evolution of the solid Earth (Creveling et al., 2012). However, the duration and magnitude of this proposed TPW and the position of South China in Rodinia are debated (Li et al., 2004; Maloof et al., 2006; Niu et al., 2016). Lack of reliable paleomagnetic results in the critical time interval is the main reason causing doubts on this TPW event. In this contribution, we conducted a detailed paleomagnetic study on the Tonian Chéngjiāng Formation red sandstone, combined with a SHRIMP II U-Pb zircon geochronological study on an interbedded tuff horizon. On the one hand, we will use these data to clarify the characteristics of this TPW event. On the other hand, using the resulting TPW paleomagnetic polar wander path, we will refine the position of continents within Rodinia, especially South China.

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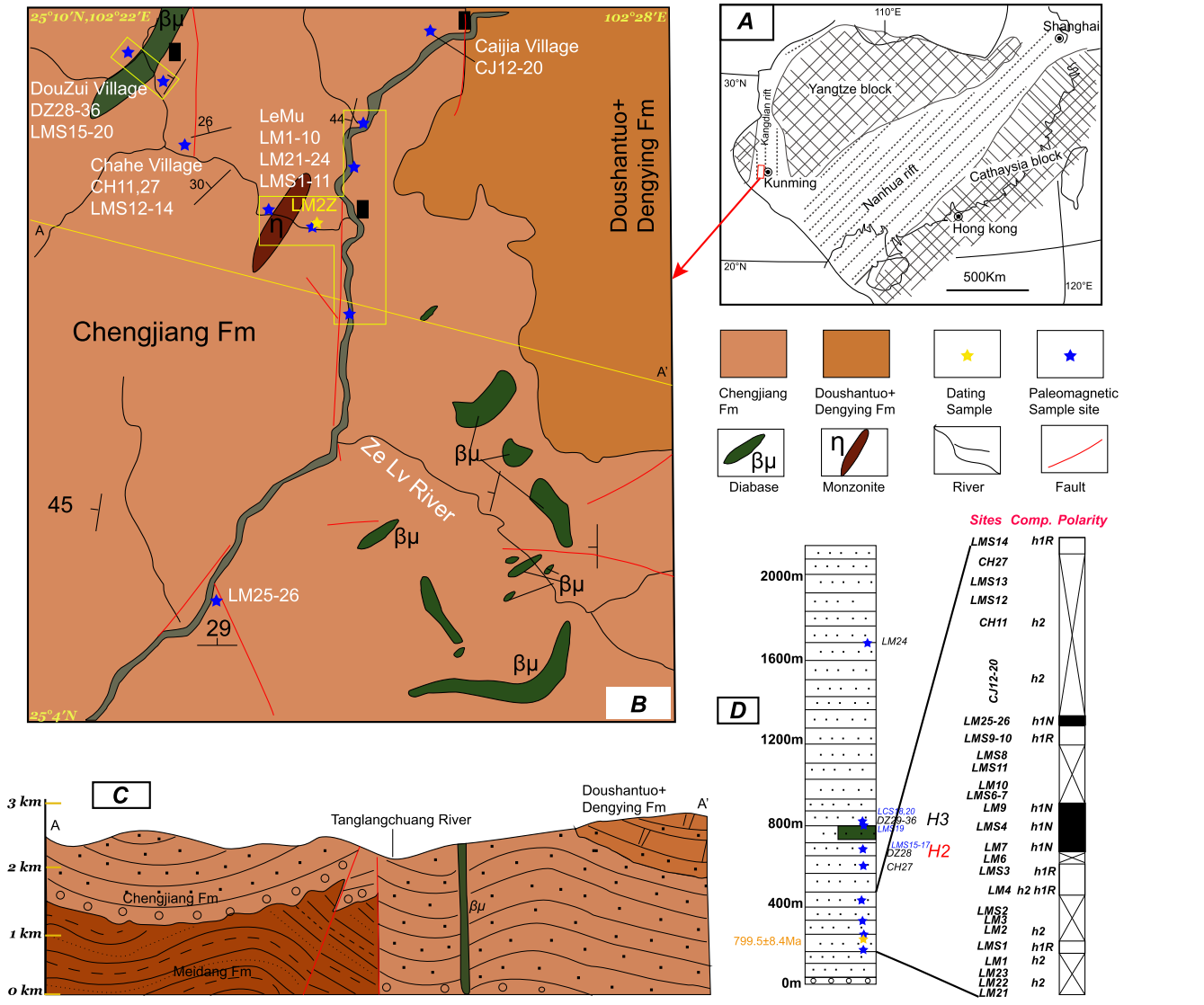


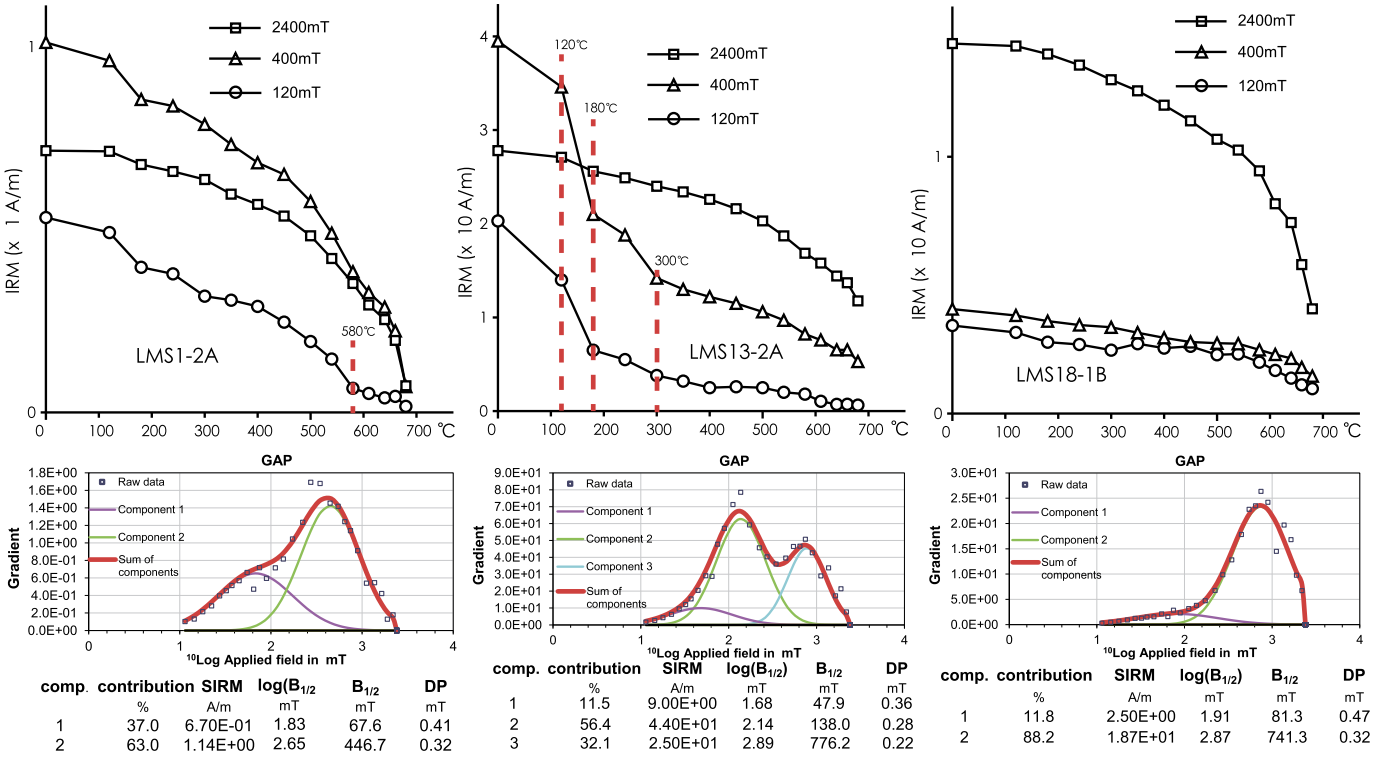
Fig. 1. (A) Simplified map of South China showing the location of the sampling area. (B) Geological sketch map of the sampling area in Xīshān district, Kūnmíng, Yúnnán Province. Four sections were sampled: Càijiā village (CJ), Lèmǔ Village (LM and LMS1-11), Chàhé village (CH and LMS12-14) and Dǒuzuǐ village (DZ and LMS15-20). (C) Measured cross section (A–A′, location shown in Fig. 1B). (D) Stratigraphic column, horizons sampled and the distribution of H1, H2, and H3 in this study; the polarity distribution is also shown.

2. Geological setting and sampling

The study area is at the southern part of the Neoproterozoic Kāngdiān Rift Basin, which is a north-south striking extensional structure developed on the Jīnníng-Sìbāo orogenic unconformity (Wang and Li, 2003, Fig. 1A). Wang and Li (2003) suggested the basin was a typical intracontinental rift basin, bordered by the Ānníngghé-Yímén and Gānluò-Xiǎojiāng basement faults on the west and east, respectively. Early basin infill comprises the Chéngjiāng Formation, which is dominated by alluvial and fluvial sediments. The sequences of the Chéngjiāng Formation consist of basal conglomerate, pebbly sandstone, medium-coarse-grained greywacke, medium-fine-grained greywacke, siltstone, argillaceous siltstone, and mudstone. The strata are dominantly purple/red in color. Volcanic activities were intense in the study area during its deposition, and thus there is a significant amount of volcanic material incorporated within the Chéngjiāng Formation; these materials formed pyroclastic rocks and occasionally interlayers of volcanics, which can be used to constrain the age of deposition. In the study area, there occur many small Variscan sills or stocks. To the east of Fùmīn county, these sills and stocks intrude early Permian strata, suggesting that they should be late Permian in age (Regional Geology of Yúnnán Province, P340-341, 1982).

Due to the India-Asia collision in the Cenozoic, there are varying degrees of vertical-axis rotation and tilting in different locations along the western edge of South China (Wang et al., 1998; Zhang et al., 2004). Although our study sites are located between the Xiānshuǐhé-Xiǎojiāng Fault (XXF) and the Yuánmóu Fault, where previous studies suggested ∼5° of rotation (Wang et al., 1998), paleomagnetic results from Éméishān large igneous province near Kūnmíng (Van Der Voo et al., 1993) have no distinguishable deviation from the results outside the XXF. This indicates that our sampling area is structurally stable, and paleomagnetic results should not be affected significantly by vertical-axis rotation (Van Der Voo et al., 1993).

We collected samples from four sections: Càijiā village (CJ), Lèmǔ village (LM), Chàhé village (CH) and Dǒuzuǐ village (DZ) in Xīshān district, Kūnmíng (Fig. 1A, B). Although sites from DZ and LM sections lie close to the Variscan stocks, all sites are separated from the stocks by at least 100 meters and show no signs of contact metamorphism. A total of 416 cores (36 sites) from a portable gasoline-powered drill, and 100 block samples (20 sites, denoted



comp.	contribution	SIRM	log($B_{1/2}$)	$B_{1/2}$	DP
	%	A/m	mT	mT	mT
1	37.0	6.70E-01	1.83	67.6	0.41
2	63.0	1.14E+00	2.65	446.7	0.32

comp.	contribution	SIRM	log($B_{1/2}$)	$B_{1/2}$	DP
	%	A/m	mT	mT	mT
1	11.5	9.00E+00	1.68	47.9	0.36
2	56.4	4.40E+01	2.14	138.0	0.28
3	32.1	2.50E+01	2.89	776.2	0.22

comp.	contribution	SIRM	log($B_{1/2}$)	$B_{1/2}$	DP
	%	A/m	mT	mT	mT
1	11.8	2.50E+00	1.91	81.3	0.47
2	88.2	1.87E+01	2.87	741.3	0.32

Fig. 2. Results of stepwise thermal demagnetization of a three-component IRM and coercivity component analysis of IRM in gradient acquisition plots (GAP). Orthogonal IRMs were imparted in fields of 0.12 T (circles), 0.4 T (triangles), and 2.4 T (squares).

by LMS), were collected. Almost all of the sites were located in the lower part of the Chéngjiāng Formation west of the river and a minor fault (Fig. 1C, D). All samples were oriented with a magnetic compass. To constrain the age of the Chéngjiāng Formation, we sampled a tuffaceous siltstone, which is located at the LM section (25°08′40″N, 102°24′57″E), at the bottom of the Chéngjiāng Formation, near sites LMS1 and LM1 (Fig. 1B, D, denoted by LM2Z).

3. Methods

All cores were cut into at least one standard specimen of 2.3 cm height and 2.54 cm diameter at Paleomagnetism Laboratory of Nanjing University. Block samples were drilled into cores and then cut into standard specimens at the Key Paleomagnetic Laboratory of the Chinese Academy of Geological Sciences, Beijing, China.

To characterize the magnetic mineralogy, representative specimens were chosen for various rock magnetic experiments. Isothermal remanent magnetization (IRM) acquisition experiments were conducted, IRMs were imparted with an ASC IM-10-30 impulse magnetizer and measured with a JR-6A spinner magnetometer. To further constrain the unblocking temperatures of the magnetic minerals, representative samples were magnetized sequentially along *Z*-, *Y*-, and *X*-axes at fields of 2.4 T, 0.4 T and 0.12 T, respectively, and subjected to stepwise thermal demagnetization (Lowrie, 1990) at the Key Paleomagnetic Laboratory of the Chinese Academy of Geological Sciences, Beijing, China.

Stepwise thermal demagnetization of pilot samples was performed using an ASC-TD48 oven to isolate the characteristic remanent magnetization (ChRM). After analyzing the behavior of the pilot samples, progressive thermal demagnetization was carried out using 14–18 steps at a 50 °C interval for low temperatures (<300 °C), and a 20–30 °C interval for high temperatures (>300 °C) up to 670 °C, and finally with a 10–15 °C interval to 685–690 °C. The remanence of the specimens was measured using a 2G Enterprises Inc. cryogenic rock magnetometer (2G-755) housed in a magnetically shielded room. The remanence directions were analyzed using principal component analysis (Kirschvink, 1980), and the site-mean directions were calculated using Fisher spherical statistics (Fisher, 1953) or using the intersections of remagnetization great-circles (McFadden and McElhinny, 1988). Paleomagnetic software packages PMGSC (by R. Enkin) and PaleoMac (Cogné, 2003) were used to perform data analyses and produce related figures. GPlates software (Boyden et al., 2011) was used to reconstruct Rodinia and produce related figures. All the magnetic measurements were made at the Paleomagnetism Laboratory of Nanjing University.

Zircons were selected and cast in an epoxy mount by handpicking under a binocular microscope together with a zircon U-Pb standard TEMORA (Black et al., 2003), and were then polished to section for analysis. Zircon grains were recorded with transmitted and reflected light micrographs and cathodoluminescence (CL) images, and the mount was then vacuum-coated with gold. Measurements were conducted using the SHRIMP II ion microprobe at the Chinese Academy of Geological Sciences. The data were processed using the SQUID and ISOPLOT programs of Ludwig (2001).

4. Results

4.1. Rock magnetic results

Cumulative Log-Gauss functions (CLG) analysis was performed on the IRM data for three representative specimens using the software developed by Kruiver et al. (2001) (Fig. 2). A low-coercivity component (47∼81 mT) that contributes 11–37% of the total IRM (Fig. 2) is present in all specimens; based on the coercivity range it is interpreted as magnetite (Peters and Dekkers, 2003). An obvious decrease of the low-coercivity component around 580–600 °C in the Lowrie test (Fig. 2) also suggests the presence of magnetite. A medium-coercivity component (138 mT) that contributes 56% of the total IRM in some specimens (Fig. 2) is interpreted as pyrrhotite, based on the coercivity range identified by Peters and Dekkers (2003). The Lowrie test also indicates

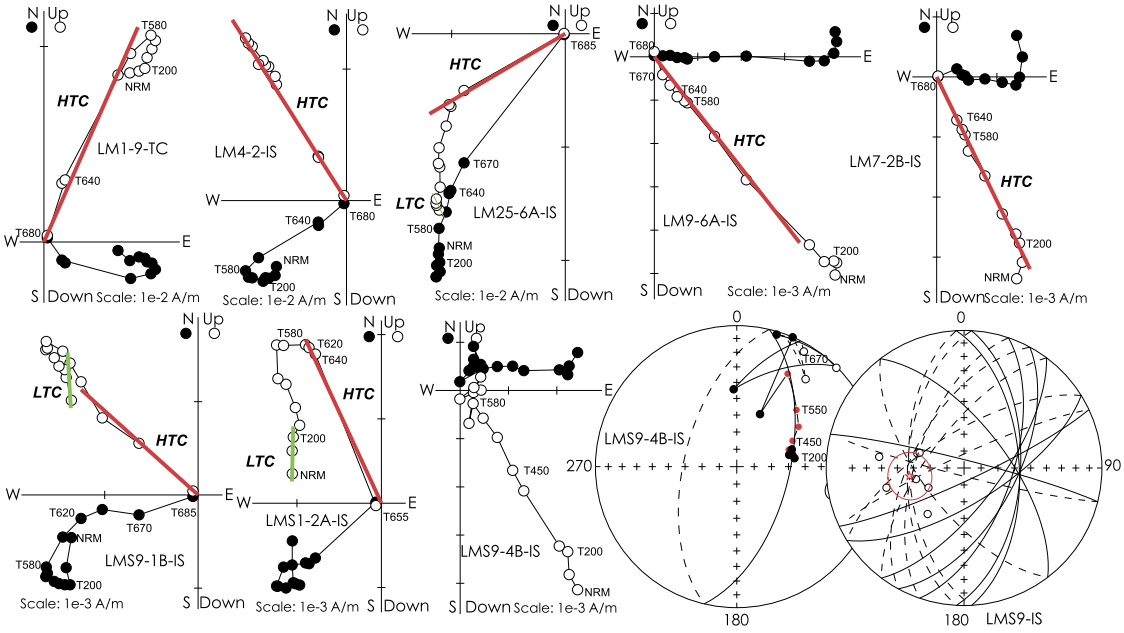


Fig. 3. Orthogonal vector projections, and their stereographic projection for samples from the Lèmǔ village section. Sample LMS9-4B shows erratic directional changes after heating to 580 °C. The site LMS9 mean direction (red-star) is calculated by the intersection of great circles method and from the directly observed components. The green line indicates the low-temperature component (LTC), and the red line the high-temperature component (HTC). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

pyrrhotite, based on the obvious decrease of the medium coercivity component at around 300 °C (Fig. 2). A high-coercivity component (440–770 mT) that contributes 32–88% of the total IRM is interpreted to be held by hematite (Peters and Dekkers, 2003). The medium- and high-coercivity components in all specimens persist up to 680 °C and then decay dramatically, indicating the presence of hematite (Fig. 2). In addition, the obvious decrease of the low-medium coercivity components in some specimens at around 120 °C indicates the presence of goethite (Fig. 2).

4.2. Paleomagnetic results

Two natural remanent magnetization components were recognized. First, a low-temperature component (LTC) was separated below 300 °C (Figs. 3, 4) in most of the samples. The mean direction of this component in all sections is $Dg = 0.4°$, $Ig = 39.7°$, $N = 43$ sites, $kg = 74.1$, $\alpha_{95} = 2.5°$ and similar to the local present field direction in geographic coordinates, and $Ds = 341.8°$, $Is = 19.8°$, $N = 43$ sites, $ks = 18.9$, $\alpha_{95} = 5.2°$ in stratigraphic coordinates (Fig. S1). The fold test (McElhinny, 1964) is negative at the 99% confidence level. Therefore, we interpret it as a recent viscous remanent magnetization. Thereafter, a high-temperature component (HTC) was separated with unblocking temperatures above 580 °C, mainly between 580 °C and 680 °C (Figs. 3, 4). However, only 25 sites reveal the HTC; other sites either only isolate the LTC or have too scattered directions for a mean direction to be meaningful. In addition, some sites demonstrate completely erratic demagnetization characteristics, which prevent the isolation of any directions. Since the HTCs are variable across the study area, the results are presented below on a section-by-section basis.

4.2.1. Lèmǔ village section (LM)

In the Lèmǔ sampling section, an almost east-west HTC is revealed up to 685 °C from 10 sites comprising 83 specimens (Fig. 3, Table 1); it is designated H1. The IRM acquisition, thermal demagnetization and thermo-remanent analyses all suggest that the remanence carrier is high-coercivity hematite. Six sites demonstrate linear decay toward the origin (Fig. 3). However, the demagnetization directions become erratic above 580 °C for LMS3 and some samples from LMS1, LMS9 and LMS10 (Fig. 3), which exhibit well-defined great circle paths at temperatures below 580 °C. The site-mean direction was then calculated using the intersection of great circles, or combined with directly observed directions, using the statistical method developed by McFadden and McElhinny (1988) (Fig. 3). Five sites demonstrate east upward directions, and five sites demonstrate west downward directions, in stratigraphic coordinates (Fig. 5B). Besides the H1, there is also an almost north-south upward direction which is mostly separated between 580 °C and 680 °C (designated by H2, Fig. 4B, E). In some cases, it appeared with the H1 component within one site (e.g. LM4, Fig. 6). As this H2 component also appears in the other three sections, we will discuss them together, below.

The average site-mean dual-polarity H1 direction is $Dg = 104.3°$, $Ig = 70.7°$, $N = 10$ sites, $kg = 7.2$, $\alpha_{95} = 19.3°$ in geographic coordinates and $Ds = 287.0°$, $Is = 69.2°$, $ks = 50.2$, $\alpha_{95} = 6.0°$ (Fig. 5A, B, Table 1) after tilt correction. The fold test of McElhinny (1964) on H1 is positive at the 99% confidence level. In addition, the directions of H1 pass the reversal test (McFadden and Lowes, 1981) at the 95% probability level (C class), and a bootstrap fold test (Tauxe and Watson, 1994) on H1 revealed that peaks in concentration were within the range of 87–145% unfolding, indicating that this component was very likely acquired before the beds were tilted.

4.2.2. Càijiā village (CJ) and Chàhé village (CH) sections

Six sites from the CJ and CH sections reveal an HTC up to 685 °C (Fig. 4A, C, D, Table 1), which is similar to the H2 component of LM section. They have an almost southward declination and upward inclination (Fig. 5D, E). Rock magnetic experiments suggest the remanence carrier is high-coercivity hematite. All sites demonstrate a linear decay to the origin (Fig. 4A, C, D).

Together with five sites with H2 component from LM section, the average site-mean direction is $Dg = 218.4°$, $Ig = -35.3°$, $N = 11$ sites, $kg = 6.7$, $\alpha_{95} = 19.0°$ in geographic coordinates, and $Ds = 193.4°$, $Is = -45.3°$, $ks = 15.8$, $\alpha_{95} = 11.8°$ in stratigraphic coordinates (Fig. 5D, E, Table 1). Although the data cluster substantially better after tilt correction (k increases more than twofold),

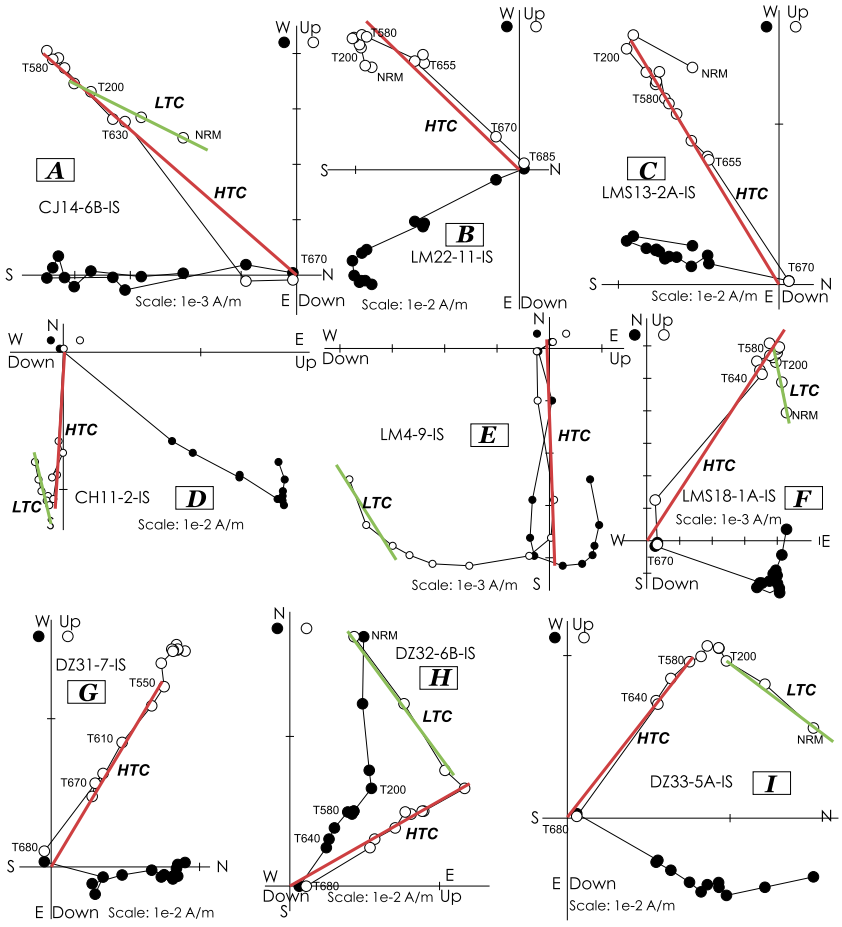


Fig. 4. Orthogonal vector projections for samples of the Càijiā village (A), Lèmǔ village (B, E), Chàhé village (C, D) and Dǒuzuǐ village (E, F, G) sections. The green line indicates the low-temperature component (LTC), and the red line the high-temperature component (HTC).

the fold test of McElhinny (1964) on H2 is inconclusive due to the low number of sites. The bootstrap fold test (Tauxe and Watson, 1994) on H2 reveals that the maximum concentrations of the data are within the range of 84–149% unfolding, which suggests that this component was acquired before the beds were tilted.

4.2.3. Dǒuzuǐ village section (DZ)

Four sites comprising 39 specimens revealed an HTC between 580 °C and 680 °C (Fig. 4F, G, H, I), designated H3. The IRM acquisitions and CLG analysis, thermal demagnetization and thermoremanent analysis all suggest that the remanence carrier is high-coercivity hematite. The direction of H3 is easterly with an upward inclination (Fig. 4F, G, H, I). The average site-mean direction is $Dg = 051.6°$, $Ig = -69.9°$, $N = 4$ sites, $kg = 46.3$, $\alpha_{95} = 13.7°$ in geographic coordinates and $Ds = 103.6°$, $Is = -46.0°$, $ks = 63.2$, $\alpha_{95} = 11.6°$ in stratigraphic coordinates (Fig. 5E, F, Table 1). The monoclinal strata precluded the application of the fold test.

4.3. Geochronologic results

Twelve SHRIMP U-Pb analyses of 12 zircon grains were obtained for sample LM2Z, and the results are listed in Table S1 and illustrated in a U-Pb concordia plot (Fig. 7). The zircon grains from this sample are euhedral-subhedral and exhibit concentric oscillatory zoning, typical of an igneous origin. The zircon grains are 30–130 μm in length, and the length/width ratios range from equant to about 3:1. Spot LM2Z-5 yielded a much younger age than the other spots (Fig. 7), which was rejected due to its high U content (Table S1). In addition, spot LM2Z-8 was excluded because of its high common ^{206}Pb (Table S1) content. The remaining 10 analyses define a concordia age of 799.5 ± 8.4 Ma ($n = 10$; MSWD = 1.4; Fig. 7).

5. Discussion

5.1. Age of the Chéngjiāng Formation and the paleomagnetic results

Previous geochronological studies (Rb-Sr whole rock dating) on the basal volcanics of the Chéngjiāng Formation in the Luoci area (Sun, 1985) indicated that the Chéngjiāng Formation was deposited at ca. 890 Ma. However, Cui et al. (2015) conducted LA-ICP-MS U-Pb dating on the basal volcanics in the same area, which yielded a weighted age of 804 ± 6 Ma. In addition, Jiang et al. (2012) reported two SHRIMP II U-Pb zircon ages of 797.8 ± 8.2 and 803.1 ± 8.7 Ma from the lower part of the Chéngjiāng Formation in the Jīnyáng and Dōngchuān areas. Together with our SHRIMP U-Pb zircon age of 799.5 ± 8.4 Ma from the bottom of the Chéngjiāng Formation (Fig. 1), we suggest that the Chéngjiāng Formation initially deposited at ~805 Ma. Dates from the middle and uppermost part of the Chéngjiāng Formation suggest that deposition ended at ~725 Ma (Cui et al., 2013).

The H1 component from the Lèmǔ village section passes both the reversal test (McFadden and Lowes, 1981) and fold tests (McElhinny, 1964; Tauxe and Watson, 1994). It records at least two and half magnetochrons (Fig. 1D), which cannot easily be explained by a secondary magnetization. In addition, the pole calculated from H1 (CJH1) is distinct from all the known Phanerozoic poles of the SCB (Fig. 8) and therefore we consider that H1 is a primary rema-

Table 1
HTC paleomagnetic data of Chéngjiāng Fm, Yúnnán.

Section	Site	Comp.	N/n	Dg	Ig	Ds	Is	k	α_{95}	Methods
Lèmǔ village (LM and LMS) (25.146°N, 102.412°E)	LMS1	H1	6	233.3	−62.1	135.9	−52.8	64.5	9.0	GC+PCA
	LMS3	H1	6	85.5	63.8	281.7	60.7	101.1	8.7	GC
	LMS4	H1	22	102.3	64.1	299.2	75.6	106.9	3.0	PCA
	LMS9	H1	16	275.2	−58.6	88.1	−66.3	17.9	9.0	GC+PCA
	LMS10	H1	5	261.2	−71.3	104.5	−63.9	39	14.1	GC+PCA
	LMS14	H1	7	32.7	−68.7	82.7	−73.4	52.7	8.4	PCA
	LM4	H1	6	243	−52.8	127.5	−64.9	21.3	14.9	PCA
	LM7	H1	4	95.1	69.4	282.7	68.8	47.7	13.4	PCA
	LM9	H1	6	98.1	55.3	305.8	76.7	92.9	7.0	PCA
	LM25	H1	5	213.8	17.2	235.4	73.2	139.4	6.5	PCA
	Mean of H1		10/83	104.3	70.7			7.2	19.3	
						287	69.2	50.2	6.9	
	Paleomagnetic poles				12.7N, 137.2E, dp/dm = 29.1/33.5 (*in situ*)					
					33.4N, 56.1E, dp/dm = 7.3/8.9 (tilt corrected)					
Lèmǔ village (LM), Chàhé village (CH) and Càijiā village (CJ) and LMS (25.146°N, 102.412°E)	LM1	H2	10	235.6	−10.9	213.1	−47.5	12.7	14.1	PCA
	LM2	H2	4	241	−16.4	221.9	−48	129.2	8.1	PCA
	LM4	H2	6	236.7	16.4	231.5	−33.6	38.4	11.0	PCA
	CH11	H2	12	221.9	−36.4	200	−40.8	25.5	8.8	PCA
	CJ14	H2	9	196.4	−53	174.8	−38.8	17.9	12.5	PCA
	CJ15	H2	9	217.8	−53	198.3	−48.4	24.5	10.6	PCA
	CJ17	H2	19	177.8	−46.7	168.4	−27.3	12.7	9.8	PCA
	CJ18	H2	3	164.7	−53	154.7	−34.6	127	11.0	PCA
	LM22	H2	6	207.3	−28	172.3	−46.3	34.1	11.6	PCA
	LMS11	H2	8	247.7	−15	217.4	−61.7	80.9	6.2	PCA
	LMS13	H2	13	201.8	−59.5	192.3	−43.1	74.2	4.8	PCA
	Mean of H2		11/99	218.4	−35.3			6.7	19.0	
						193.4	−45.3	15.8	11.8	
	Paleomagnetic poles				54.2N, 193.6E, dp/dm = 12.7/21.9 (*in situ*)					
					77.8N, 191.9E, dp/dm = 9.5/15.0 (tilt corrected)					
Dǒuzuǐ village (DZ) (25.168°N, 102.385°E)	LMS18	H3	6	94.6	−67.9	112.3	−34.1	26.8	13.7	GC+PCA
	DZ31	H3	13	54.6	−70	100.3	−46.7	13.3	11.8	PCA
	DZ32	H3	8	40.9	−71.1	106.8	−49	39.1	9.0	PCA
	DZ33	H3	12	24.5	−62	92	−53.2	19.2	10.2	PCA
	Mean of H3		4/39	51.6	−69.9			46.3	13.7	
						103.6	−46	63.2	11.6	
	Paleomagnetic poles				0.6N, 74.8E, dp/dm = 20.2/23.5 (*in situ*)					
					22.6N, 33.2E, dp/dm = 9.5/14.8 (tilt corrected)					

Comp. = Component; N, number of sites for statistical analysis; n, number of samples for statistical analysis; Dg, Ig, Ds, Is, declination and inclination in geographic and stratigraphic coordinates; k, Fisher precision parameter of the mean; α_{95}, confidence of the mean direction; dp/dm, semi-axes of elliptical error around the pole at a probability of 95%. GC, Great circle intersection; PCA, Principal component analysis.

nence of the Chéngjiāng Formation. Since all sites containing the H1 component are either near the sampling horizon of LM2Z or from the lower part of the Chéngjiāng Formation (Fig. 1D), we assign an age of ∼800 Ma to the H1 component.

A positive fold test (Tauxe and Watson, 1994) on H2 component only constrains its acquisition before Cenozoic time. In addition, the tilt-corrected pole calculated from H2 (CJH2) overlaps with the mean Late Cretaceous pole for South China (Fig. 8). It is noteworthy that the rock magnetic results indicate that samples with H2 contain a more complex assemblage of magnetic carriers, including magnetite, pyrrhotite, hematite and goethite (Fig. 2, LMS13-2A), which may reflect complex magnetization processes. Actually, samples from LM4 with H1 and H2 components demonstrate different demagnetization characteristics (Fig. 6). While samples with H1 have typical detrital-remanent magnetization (DRM) thermal demagnetization curves, samples with H2 demonstrate chemical-remanent magnetization (CRM) thermal demagnetization curves (Fig. 6) (Jiang et al., 2015). In addition, many studies have reported severe Jurassic-recent remagnetization of red-beds in Southwest China (Yúnnán, Gùizhōu Provinces) (Gao et al., 2018; Huang and Opdyke, 2015). Therefore, considering the regional geological setting (the India-Asia collision since the Cenozoic), we suggest that the H2 component is a (pre-folding) Late Cretaceous secondary remanence.

Due to its single polarity and nearly identical bedding attitude among relevant sites, neither the reversal test nor the fold test can be conducted on the H3 component. However, the pole calculated from H3 (CJH3) is distinct from known Phanerozoic poles of the SCB (Fig. 8), but overlaps with the 806 Ma Pole-B of Yánbiān dykes (Niu et al., 2016) and the 802 Ma pole of Xiǎofēng dykes (XFD, Li et al., 2004) in stratigraphic and geographic coordinates, respectively. Although no direct dating from the Dǒuzuǐ village section is available, Lu et al. (2013) reported a SHRIMP U-Pb zircon age of 785 ± 12 Ma from the middle part of the Chéngjiāng Formation. In addition, given the similarity between the H3 poles and the ∼800 Ma poles of the SCB, H3 is interpreted as a probably primary remanence acquired during deposition of the middle-upper Chéngjiāng Formation. However, due to the limited number of specimens, we regard it as a preliminary result, and more work is needed to confirm it.

5.2. True Polar Wander (TPW)

Previous studies suggested a possible TPW event that was initiated around ∼825 Ma (Li et al., 2004; Maloof et al., 2006; Niu et al., 2016). However, its duration and behavior were debated (Li et al., 2004; Maloof et al., 2006; Niu et al., 2016). Li et al. (2004) first proposed it was a ∼90° TPW event during ∼800–750 Ma. Considering that the Liántuó Formation pole used

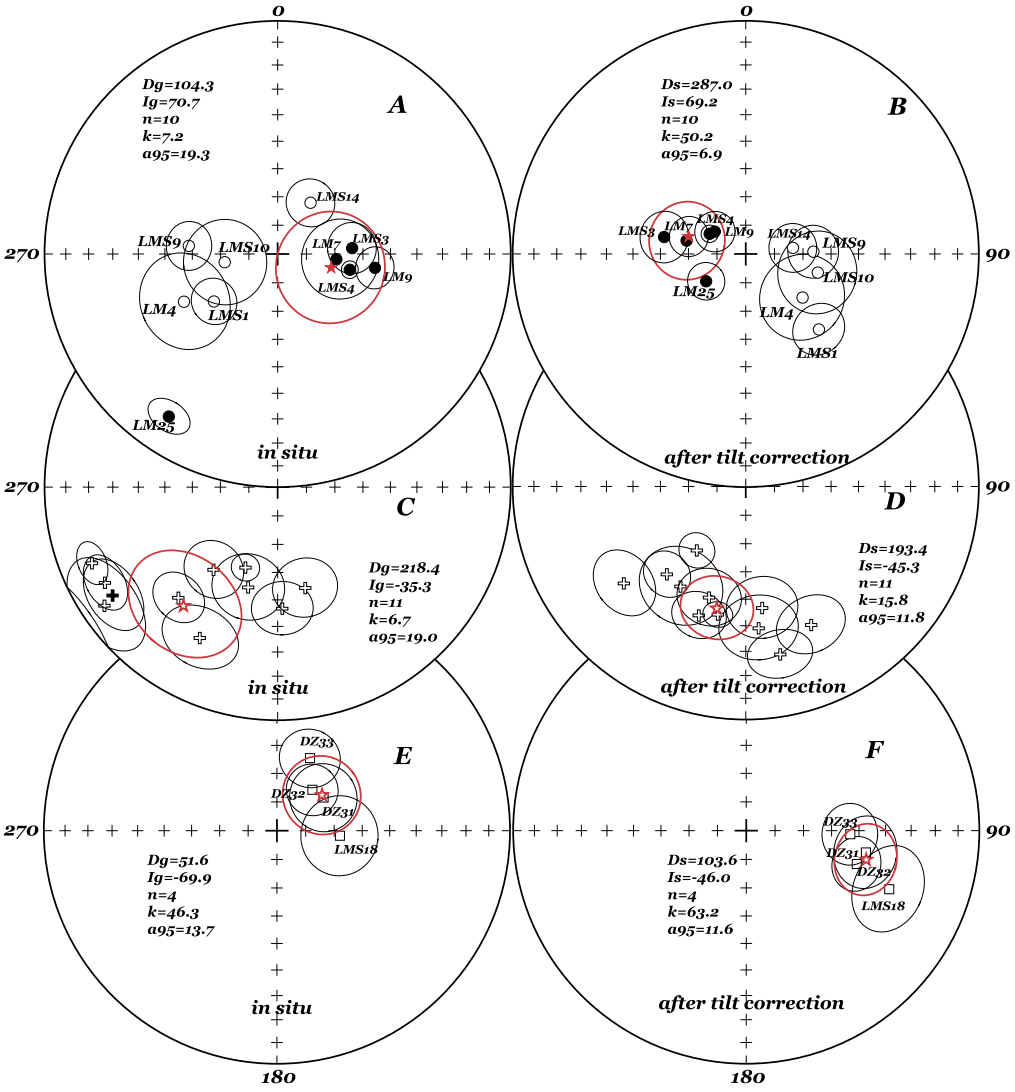


Fig. 5. Equal area plots of H1, H2 and H3 in geographic (A, D, G) and stratigraphic (B, E, H) coordinates. Red star represents the site mean direction.

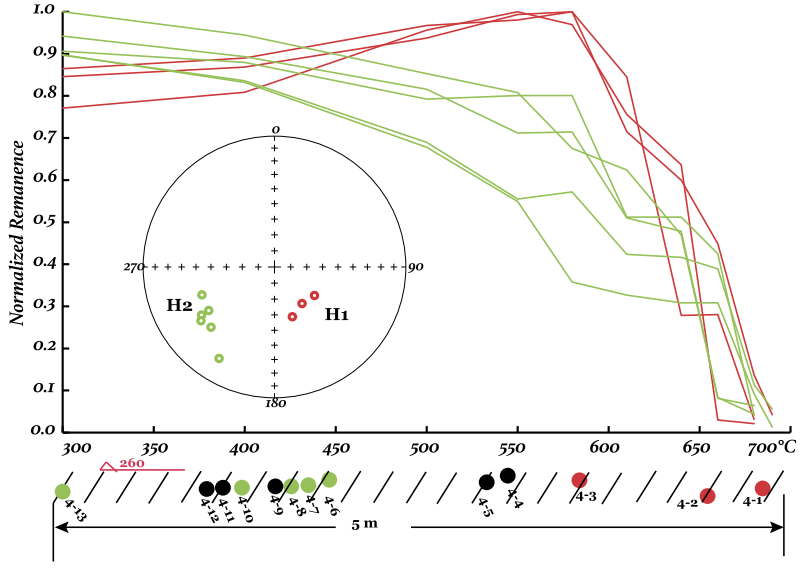


Fig. 6. Normalized thermal demagnetization remanence curves of LM4 samples and their high temperature components in stereographic plot. The red and green ones are interpreted as H1 and H2, respectively. The details about their distribution in the section are also showed in the bottom.

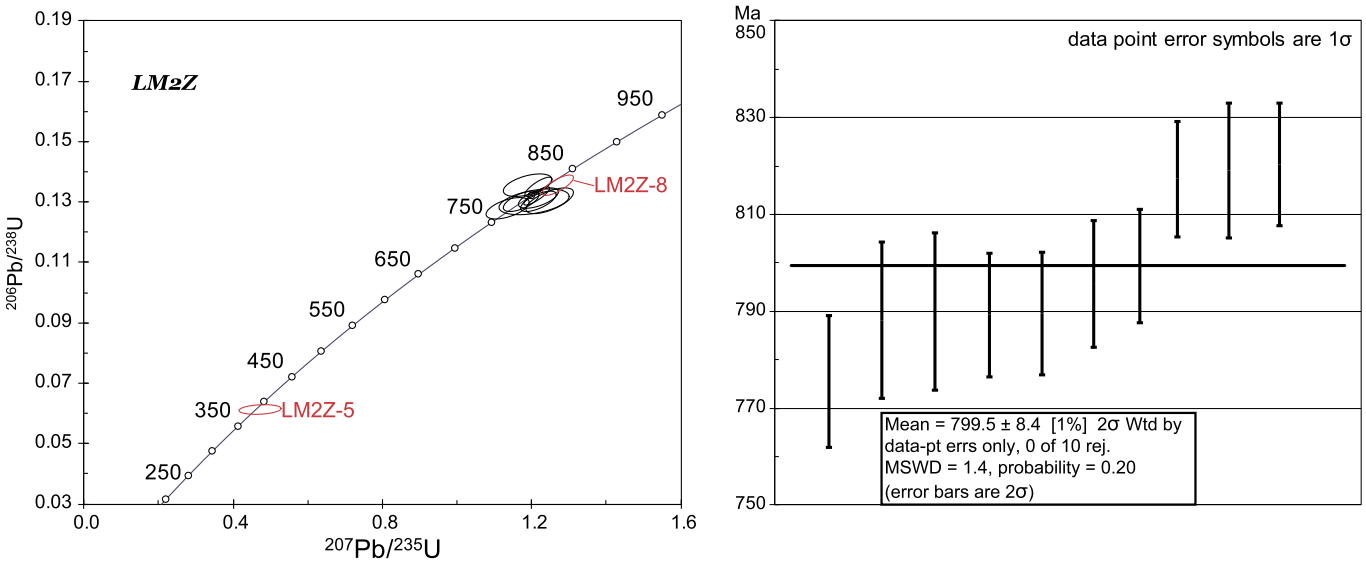


Fig. 7. Concordia plots of SHRIMP U-Pb zircon analyses and weighted average $^{206}Pb/^{238}U$ age of tuffaceous siltstone sample LM2Z from the bottom of the Chéngjiāng Formation. Spots LM2Z-5, 8 are rejected because of their unusual U or Pb content.

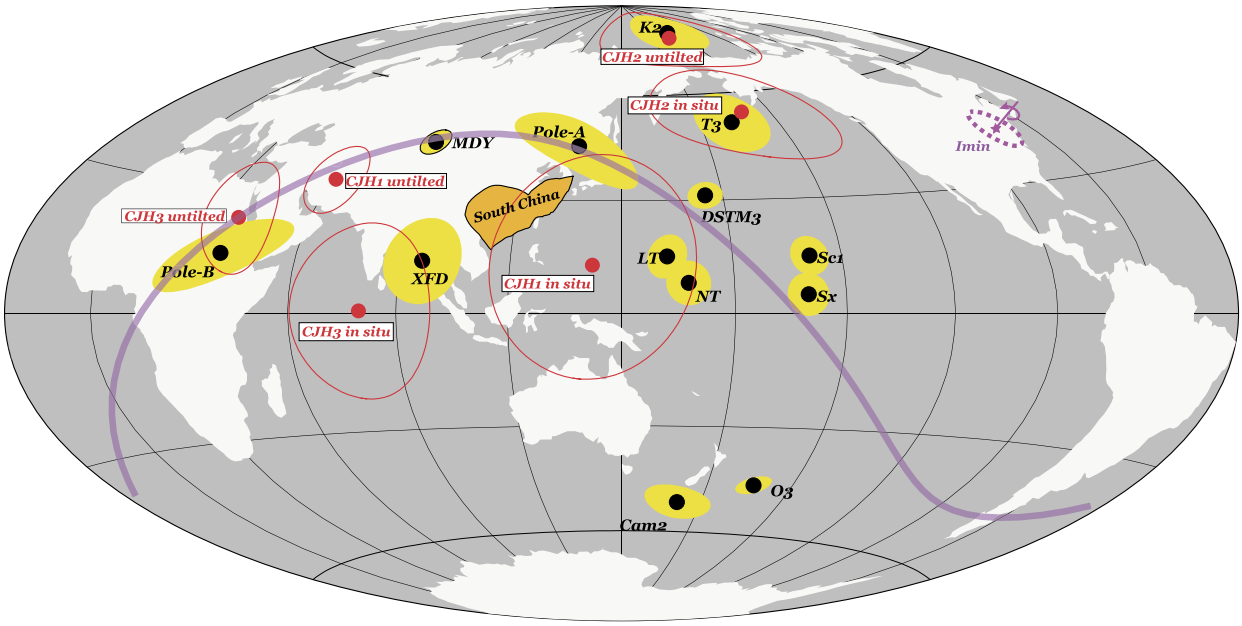


Fig. 8. Plot of Paleomagnetic poles calculated from H1, H2 and H3 and the existing Neoproterozoic and Phanerozoic paleopoles (yellow shading) from the South China block. A great circle (purple) is also fitted to Pole-A, MDY, CJH1, CJH3 and Pole-B, with corresponding minimum inertial axis (I_{min}) implied by a TPW interpretation of those results. Pole-A and Pole-B from Yánbiān dykes (Niu et al., 2016), XFD = Xiǎofēng dykes (Li et al., 2004), MDY = Mǎdǐyì Formation (Xian et al., 2017), LT = Liántuó Formation, NT = Nántuó Formation, DSTM3 = Dǒushāntuó Member3, Cam2 = Middle Cambrian, O3 = Late Ordovician, Sx = Early Silurian, Sc1 = Middle-Late Silurian, T3 = Late Triassic, K2 = Late Cretaceous. For pole references not listed above, see Fig. 16 in Jing et al. (2018).

in their study is now likely to be younger than 730 Ma (Jing et al., 2015), this unusually long time (ca. 70 million years) TPW event may be a composite of TPW and plate motion. Later, Maloof et al. (2006) suggested a transitory TPW event (from 805 to 790 Ma) based on a paleomagnetic study of the strata of the Bitter Springs carbon-isotopic excursion; however, at the time, the age of the Bitter Springs interval was not well constrained. Recently, Niu et al. (2016) suggested that this TPW event occurred during ~825–750 Ma based on their paleomagnetic and geochronologic studies of the Yánbiān dykes. However, the TPW great circle that was fitted always excluded two poles of vital time interval (see their Fig. 10). Overall, Maloof et al. (2006) suggested that this TPW event consisted of two 55° shifts; however, Li et al. (2004) and Niu et al. (2016) considered that it mainly constituted a single 90–110° polar wander shift.

Recently, an additional reliable pole from the SCB was reported: a ~810 Ma pole from the Mǎdǐyì (MDY) Formation red beds with dual polarity and a positive fold test (Xian et al., 2017), lying midway between the "A" and "B" poles from Yánbiān dykes (Niu et al., 2016). Considered together, the Yánbiān dykes poles "A" and "B", MDY, and the two tilt-corrected (presumably primary) Chéngjiāng H1 and H3 poles from this study, all lie on the same great circle (Fig. 8), conforming to the TPW hypothesis of insignificant SCB drift relative to a fixed minimum inertial axis through 825–790 Ma.

Within that framework, the ca. 802 Ma Xiǎofēng dykes pole (XFD) lies away from the great circle of other coeval poles. Al-

though all previous researchers regarded the XFD pole as reliable for the SCB, some uncertainty still remains for the following reasons. First, the geochronological age was obtained from a felsic dyke which was later cut by more mafic dykes, while the paleomagnetic results were mainly obtained from the more mafic dyke samples. Thus, the XFD pole may be younger than the zircon age. Second, the XFD pole is located far away from the almost coeval CHJ1 pole (Fig. 8). Closer inspection of the Xiǎofēng dataset shows that it can be divided into two spatially defined subgroups of sites separated by a fault (Fig. 1b in Li et al., 2004; Fig. S2): we refer to the sites south of the fault as Part-1 and north of the fault as Part-2. Results from the Part-1 subset have a declination more easterly while those from Part-2 are more northerly (Fig. S2). Because Part-2 is between two faults, results from this region may experience some local rotation. The deviation of strikes of these two parts may reflect the local counter-clockwise rotation of Part-2 (Fig. S2). Alternatively, or in combination, Part-2 might have experienced a small amount of Northwest-down tilting between the bounding faults. In this study, we recalculate a paleopole of Xiǎofēng dykes (XFD-1) only using the results from Part-1 (Table 2). This new pole is much closer to the CJH1 and MDY poles (Fig. 9).

Maloof et al. (2006) reported three paleopoles about 800 Ma from the lower Grusdievbreen Formation (lGfm), the upper Grusdievbreen Formation (uGfm), and the Svanbergfjellet 4 Member (S4mb) from East Svalbard. The occurrence of dual polarity in the lGfm and uGfm sections, and positive fold tests, suggest that the paleomagnetic data from these two sections are primary. A successful synsedimentary fold test at the 99% confidence level suggests that the magnetization of S4mb section was acquired during sedimentation or during earliest diagenesis (Maloof et al., 2006). A recent geochronologic and chemostratigraphic study of the Bitter Springs isotopic stage (Swanson-Hysell et al., 2015) constrains the ages of these three poles to ∼810 Ma, ∼805 Ma and ∼790 Ma, respectively.

As distinct from calculating post-Mesozoic TPW, which can be measured by the paleomagnetic results and hotspot tracks, recognition and estimation of pre-Mesozoic TPW is dependent on its occurrence at much faster rates (Evans, 2003). In this case, APWPs of all continents should look similar or distribute along a single great circle (after restoration of the continents into the correct paleogeography; Kirschvink et al., 1997).

First, we calculated the APW rate of South China and Svalbard, respectively. The calculation shows that South China moved extremely rapidly with a speed over ∼2.27°/Myr or ∼25 cm/yr during 825–800 Ma. Through a similar time interval (810–805 Ma), the speed of Svalbard was much higher than South China, ∼9.02°/Myr or ∼100 cm/yr; and after that it slowed down but still was faster than South China, with a speed of ∼7.33°/Myr or 81 cm/yr until 790 Ma. The differences of speed between South China and Svalbard may mainly be due to the age uncertainty of both sets of poles. Overall, speed of South China and Svalbard during 825–795 Ma clearly exceeds the theoretical (∼20 cm/yr) and the currently observed fastest plate speed (8–10 cm/yr) on Earth (Lallemand et al., 2005; Zahirovic et al., 2015). Secondly, APWPs of South China and Svalbard show similar features. They all moved first to one direction and then changed direction at about 800 Ma (Fig. 9A). By restoring the APWP of Svalbard to that of South China (Euler rotation = 32.6°, 162.4°, +144°), lGfm, uGfm and S4mb poles from Svalbard overlap the Pole-A, CJH1 and XFD-1 poles, and a great circle can be fitted to these poles with most of them along it (Fig. 9A). Due to its fast speed, great circle distribution, short duration and large angular distance (∼63°), the combined APW path can be well explained by a Type-II TPW model (Evans, 2003; Raub et al., 2007). Relative sea-level and $\delta^{13}C$ records reported by Maloof et al. (2006) for Svalbard also supported the TPW model, which we will not reiterate here.

5.3. Magnetochrons

As the CJH1 pole is compared with the uGfm pole of Svalbard in our reconstruction (Fig. 9A), and the fact that both of those poles have dual polarity, there is potential for magnetostratigraphic correlation between the two widely separated regions, and a consistency test of the reconstruction itself. In Svalbard (Maloof et al., 2006), the uGfm dataset is dominated by southeasterly-directed ChRMs; although the text of Maloof et al. (2006) assigned "normal" absolute geomagnetic polarity to those vectors, the APWP illustrated in that paper (as well as our own reconstructions in Fig. 9) would be most feasibly integrated into Neoproterozoic kinematic models (e.g., Li et al., 2008, 2013; Merdith et al., 2017a) if the dominant southeasterly-directed uGfm data had "reversed" absolute geomagnetic polarity. In that context, our new CHJ1 dataset has a slight dominance of southeasterly-up ChRM directions, which we also interpret as "reversed" geomagnetic polarity. Though our composite stratigraphic column is crude relative to the sub-meter-scale sampling of Maloof et al. (2006), both studies show the dominant polarity of remanence ("reversed" in our interpretation) enclosing two relatively narrow bands of the minority polarity ("normal" in our interpretation). Thus, our relative reconstruction model of South China and Svalbard is broadly consistent with the available stratigraphic constraints from the two studies. Further magnetostratigraphic study of both regions could test whether additional polarity zones might be present in either area.

5.4. Implications for the reconstruction of Rodinia

Previous Rodinia reconstructions are mainly based on geological evidence (Ernst et al., 2008; Goodge et al., 2008; Karlstrom et al., 1999; Li et al., 1995; Moores, 1991), or on representative "key" poles of each continent or terrane (Evans et al., 2000, 2016; Wingate et al., 2002), which are non-unique in terms of their relative position to each other. Consequently, the configuration of the Rodinia is debated (Evans et al., 2009, 2016; Jing et al., 2015; Karlstrom et al., 1999; Li et al., 1995, 2008; Merdith et al., 2017a, 2017b; Wen et al., 2018; Wingate et al., 2002). However, during a large and rapid TPW event, all continents (no matter whether parts of a supercontinent or separated fragments) should display similar polar wander paths (Van der Voo, 1994; Kirschvink et al., 1997). The common path, therefore, can be used to restore the continents both in latitude and the relative paleolongitude (Evans, 2003). Such TPW-based reconstruction has been viably implemented in Ediacaran and Phanerozoic paleogeography (Kirschvink et al., 1997; Mitchell et al., 2012; Raub et al., 2007; Robert et al., 2017; Van der Voo, 1994). In this contribution, the reconstruction of Rodinia is revised based mainly on the reliable 920–720 Ma paleomagnetic poles with $Q \geq 4$ (Van der Voo, 1990) from major Precambrian continental blocks (Table 2) in the TPW paleomagnetic frame we recognize above.

Laurentia is restored to its position by rotating its ∼770 ± 30 Ma Uinta Mountain Group pole (UMG, Weil et al., 2006) and the ∼780 Ma Gunbarrel mafic dykes pole (GMD, Harlan et al., 2008) close to the CJH1 pole and ∼790 Ma S4mb pole (Fig. 9B). This reconstruction locates Svalbard near the north-east corner of the Greenland (Fig. 9C, D), which was also suggested by Gee and Teben'kov (2004). The reconstruction also clearly demonstrates that South China was far distant from the Laurentia and cannot be a "missing-link" block between Australia and Laurentia at ca. 800 Ma.

In previous models, Australia-Mawson block has been located in different positions relative to western Laurentia, e.g. SWEAT

Table 2
Summary of late Tonian paleomagnetic poles from major Precambrian blocks or terranes (with quality index $Q \geq 4$, Van der Voo, 1990).

Stratigraphy	Abbreviation	Petrology	Age	Location	D	I	n	α_{95}	λ	Φ	dp	dm	Q							Drawback	References
													1	2	3	4	5	6	7		
East Svalbard																					
Svanbergfjellet 4 Mb	S4mb	Limestone	790	78.5N, 18E	333.2	29.4	18	7	−25.9	46.8	4.3	7.7	y	n	y	y	y	n	y	SV	***Van der Voo, 1990*** ***Maloof et al., 2006;*** ***Swanson-Hysell et al., 2015***
U. Grusdievbreen Fm	uGfm	Limestone	805	79.2N, 18.38E	126.9	15.9	107	2.7	1.6	71.9	1.4	2.8	y	y	y	n	y	y	y	SV	
L. Grusdievbreen Fm	lGfm	Limestone	810	79.24N, 18.2E	348.3	17.3	53	4.1	−19.6	31.3	2.2	4.2	y	y	y	n	y	y	y	SV	
South China																					
Xiǎofēng Dykes	XFD	Felsic dykes	802 ± 10	30.96N, 111.23E	51.3	−76.6	54	6.1	13.5	91	10.5	11.3	y	y	y	n	y	y	y	Sv and age	***Li et al., 2004***
Xiǎofēng Dykes	XFD-1	Felsic dykes	802 ± 10						26.1	82.1	14.0	15.2	y	y	y	n	y	y	y		***recalculated after Li et al., 2004***
Yánbiān dyke	PoleA	Mafic dykes	824 ± 6	26.886N, 101.546					45.1	130.4	19.0	19.0	y	y	y	y	n	y	y		***Niu et al., 2016***
Mǎdǐyì Fm	MDY	Red sand	810		312.5	75.7	100	4.7	44.8	80.2	3.4	3.4	y	y	y	y	y	y	y		***Xian et al., 2017***
Chéngjiāng Fm	CJH1	Red sand	800	25.145N, 102.411E	111.9	−66.7	72	5.2	32.8	56.3	7.1	8.6	y	y	y	y	y	y	y		***this study***
Chéngjiāng Fm	CJH3	Red sand	785	25.162N, 102.424E	104.9	−46.5	32	16.3	23.8	33.2	13.5	20.9	n	n	y	n	y	n	y	SV	
Liántuó Fm	LT	Red sand	720	111.069E, 30.819N	103.4	64.1	150	3.7	13.2	155.2	5.3	5.3	y	y	y	n	y	y	y		***Jing et al., 2015***
Tarim																					
Qiaoenbrak Fm	QEB	Sandstone	730	40.923N, 79.461E	246.1	30.4	81	11	−6.3	17.5	9.1	9.1	n	y	y	y	y	y	y		***Wen et al., 2013***
Baiyisi volcanic rocks	BYSV	Volcanic rocks	755 ± 15	41.6N, 86.54E	245.2	1.7	53	6	−17.7	14.2	3.0	6.8	y	y	y	n	y	n	y		***Huang et al., 2005***
Sailajiazitage Group	SLJZTG	Volcanic rocks	880 ± 10						−23.5	37	11.3	11.3	y	y	y	y	y	n	y		***Wen et al., 2018***
Australia																					
Mundine Well dyke swarm	MDS	Mafic dykes	755 ± 3				88		45.3	135.4	4.1	4.1	y	y	y	y	y	n	n		***Wingate and Giddings, 2000***
Johnny's creek Mb	JCM	Red siltstone	770 ± 10	Amadeus basin	110	−40.9	116	11.2	15.8	83	13.5	13.5	n	y	y	y	y	n	y		***Swanson-Hysell et al., 2012***
Hussar Fm	Hf	Sandstone	800-760						62.2	85.8	7.3	14.6	y	y	y	n	y	y	y		***Pisarevsky et al., 2007***
India																					
Malani Igneous Province	MIS	Felsic dykes	750	Rajasthan, NW India	2.9	63.7	272	4.2	69.4	75.7	6.5	6.5	y	y	y	y	y	y	y		***Meert et al., 2013***
São Franscisco/Congo																					
Bahia coastal A-Normal	BCAn	Mafic dykes	920						−14	288	7.0	7.0	y	y	y	y	y	n	n		***Evans et al., 2016***
Laurentia																					
Nankoweap Fm	Nf	Red sandstone	900 ± 50						−10	163	3.5	9.3	n	y	y	y	y	y	y		**Weil et al., 2003**
Gunbarrel mafic dykes	GMD	Mafic dykes	780	Southwest Montana					9.2	138.7	9.0	9.0	y	y	y	n	y	n	y		***Harlan et al., 2008***
Uinta Mountain Group	UMG	Red sandstone	770 ± 30						0.8	161.3	4.6	4.6	n	y	y	n	y	y	n		***Weil et al., 2006***
Kwagunt Fm	Kf	Red sandstone	>740						18.2	166	5.2	9.4	n	y	y	y	y	y	n		***Weil et al., 2004***
Franklin igneous rocks	FLIP	Mafic dykes	720						8.4	163.8	2.8	2.8	y	y	y	y	y	y	n		***Denyszyn et al., 2009***

1–7 and Q, reliability criteria of Van der Voo (1990). 1—Well-dated rock age, 2—Sufficient number of samples, 3—Adequate demagnetization, 4—Field tests, 5—Structural control and tectonic coherence with craton or block involved, 6—Presence of reversals, 7—No resemblance to paleopoles of younger age. λ, latitude of the paleopoles; Φ, longitude of the paleopoles; dp/dm, semi-axes of elliptical error around the pole at 95% probability.

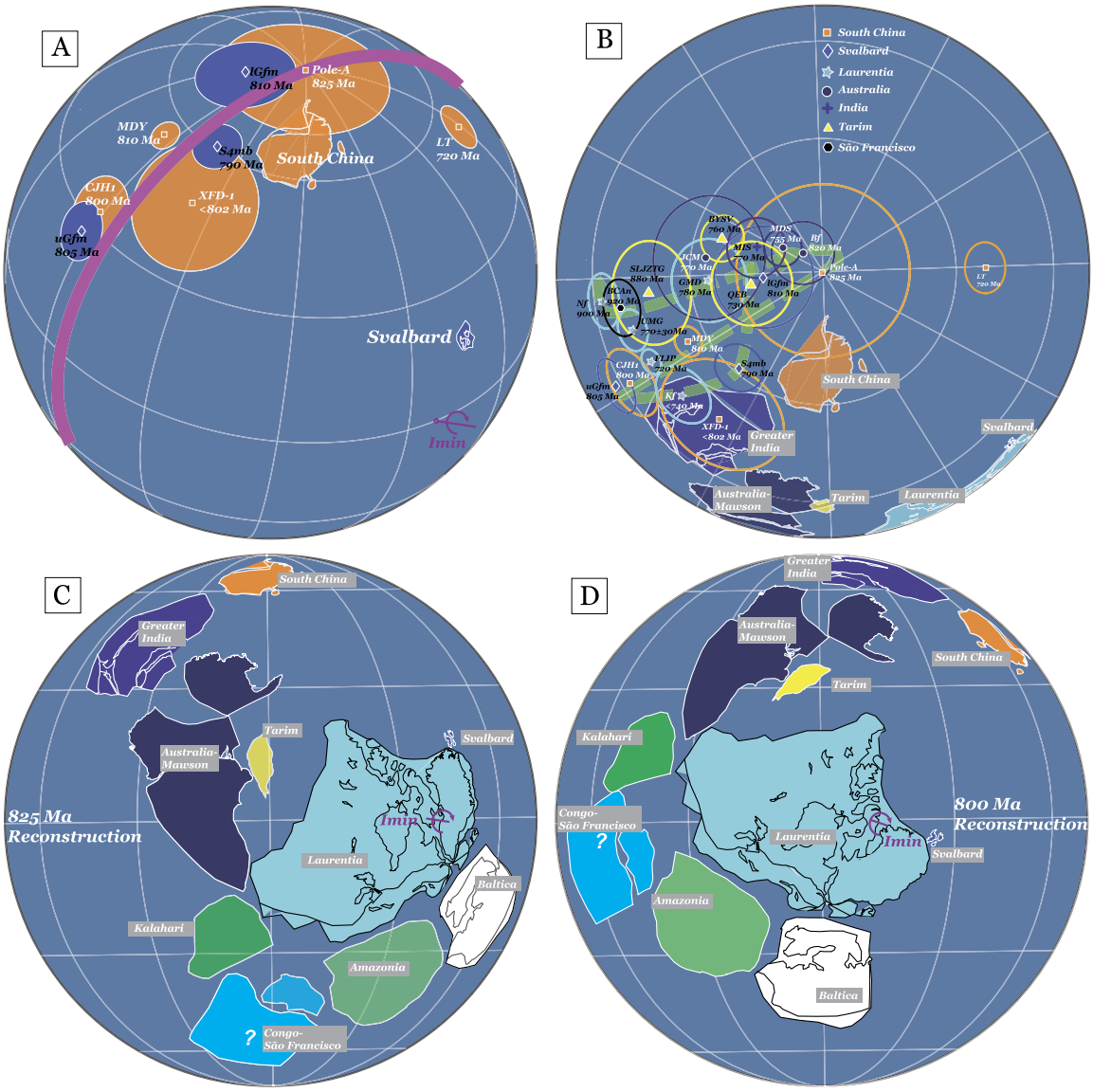


Fig. 9. (A) 825 Ma reconstruction of South China and Svalbard. Svalbard is first restored to South China by a Euler rotation of 32.6°N, 162.4°E, +144°CCW; then the ~825 Ma Pole-A of South China is restored to the north pole using a Euler rotation of 4.0°N, 41.8°E, 45.1°. A great circle (purple) is fitted to the ~825–790 Ma poles of both blocks. (B) APWPs comparison of main continents based on the TPW frame. (C) Paleomagnetically-based paleogeographic reconstruction of Rodinia at 825 Ma. Position of Baltica and Amazonia are according to Gong et al. (2018) and Cawood and Pisarevsky (2017). (D) Paleomagnetically-based paleogeographic reconstruction of Rodinia at 800 Ma.

(Moores, 1991), AUSWUS (Karlstrom et al., 1999; Burrett and Berry, 2000) and AUSMEX (Wingate et al., 2002). In our reconstruction, we restore the Australia-Mawson block to a position similar to the "Missing-Link" model (Li et al., 1995, 2008) by overlapping the 830–800 Ma Browne Fm pole (Bf, Pisarevsky et al., 2007) and ~770 Ma Johnny's Creek Member pole (JCM, Swanson-Hysell et al., 2012) of Australia with the 824 Ma Pole-A of South China and the ~780 Ma GMD pole of Laurentia, respectively (Fig. 9B). The "missing link" between Australia-Mawson and Laurentia recently was filled by Tarim and neighboring blocks in central Asia, by Wen et al. (2018). Tarim is restored adjacent to southeast Australia by overlapping its newly reported ca. 880 Ma Sailajiazitage Group pole (SLJZTG, Wen et al., 2018) with the ca. 900 Ma Nankoweap Fm pole (Nf, Weil et al., 2003) of Laurentia; and 770–730 Ma Baiyisi volcanic rocks pole (BYSV, Huang et al., 2005) with the ~780 Ma GMD pole of Laurentia and ~770 Ma JCM pole of Australia. The rest of the available space between Australia-Mawson and Laurentia may be occupied by Karakum, Qaidam-Kunlun and perhaps North China (Fu et al., 2015; Wen et al., 2018).

We overlap the 770–750 Ma Malani igneous rocks pole (MIS, Meert et al., 2013) with the ~755 Ma Mundine Well dykes swarm pole of Australia (MDS, Wingate and Giddings, 2000) and the BYSV pole, in a composite reconstruction that locates India between South China and north Australia (Fig. 9B, C, D). This reconstruction is broadly consistent with the proposed India-South China connection during the Neoproterozoic by Jiang et al. (2003) based on a comparison of Neoproterozoic stratigraphic facies assemblages from both continents. Similar tectonothermal events recorded in Afghanistan and South China (Merdith et al., 2017a) also suggest that South China probably located to the north of India. In addition, both India and the Seychelles have Neoproterozoic continental arc magmatism that is similar to that in Australia and South China in terms of age distribution (Ashwal et al., 2002), all of which supports the conclusion that they were close to each other.

In our reconstruction, the ~750 Ma poles from Australia, Tarim and India are close to each other (Fig. 9B), which suggests that a large portion of Rodinia may still have been consolidated until 750 Ma. However, the ~720 Ma poles from Laurentia, South China and Tarim are distant from each other (Fig. 9B), which suggests

that Rodinia may have broken apart during 750–720 Ma. This is consistent with the intensive bimodal magmatism during 780–750 Ma in South China, Australia, Laurentia, India and Tarim (Li et al., 2003; Lu et al., 2008).

Over the interval investigated here (900–750 Ma), the restored paleopoles appear to be clustered in largely two stable pole positions, with few intermediate directions (Fig. 9B). This "stroboscopic effect" is consistent with non-linear TPW as discussed by Mitchell (2014), which conforms to the TPW interpretation of our results and the reconstruction. On the whole, our positioning of the TPW rotation axis (minimum moment of inertia, or I_{min}) relative to central cratons within the Rodinian landmass (e.g., Laurentia) is closely similar to that of some previous reconstruction models that incorporate oscillatory TPW during Tonian time (Maloof et al., 2006; Mitchell et al., 2012), or broadly similar to others (Niu et al., 2016).

Wen et al. (2018) proposed that Rodinia finally assembled at ca. 900 Ma through a mega-dextral shearing along the Tarimian orogen, similar to the collision scenario that created Pangea (Domeier and Torsvik, 2014). Traditionally, Rodinia was treated as a supercontinent distributed across the tropical zone (Li et al., 2008). However, our reconstruction demonstrates that at least the South China and India sectors were located near or within the polar zone during ~880–750 Ma (Fig. 9C, D). Meanwhile, Australia, Baltica, Amazonia and Congo (Evans et al., 2016; Gong et al., 2018) occupied the middle to high latitude zone in both hemispheres (Fig. 9C, D). Massive late Tonian red bed and evaporites deposition and no glacial activity in these higher-latitude continents suggest a greenhouse climate during that period (Schmid, 2017). The pan-latitudinal geographic distribution through its amalgamation period makes Rodinia a more Pangea-like supercontinent, stretching to both polar zones. However, the relative climatic effects of supercontinental paleogeography differ slightly between Rodinia and Pangea. As noted above, Rodinia assembled and existed during a warm interval, but the planet plunged into global glaciation immediately following breakup (Goddéris et al., 2003). In contrast, Pangea assembled during cold climates of the late Paleozoic, existed during a warm period, and fragmented during the cooler but ice-free early Mesozoic (Royer, 2006). These differences between paleoclimates of Rodinia and Pangea suggest that some crucial impacts of supercontinental paleogeography and the Earth's system remain to be discovered.

6. Conclusions

A reliable ~800 Ma paleopole documenting at least five polarity chrons is revealed through paleomagnetic and geochronological study on the Tonian Chéngjiāng Formation red-beds from South China. Combined with previous paleomagnetic results from Svalbard and South China, we support the existence of a true polar wander (TPW) event during the late Tonian Period, approximately from 825 Ma to 790 Ma, with a one-way angular distance of 63°. Under this TPW paleomagnetic reference frame, we reconstruct the (paleo)northern portion of Rodinia by using the reliable late Tonian paleomagnetic results from several cratons. In this reconstruction, South China is distant from Laurentia and Australia, and thus was not the key block in the "missing-link" model between those cratons. Instead, following earlier suggestions, Tarim occupies the position between Australia and Laurentia. After 750 Ma, poles from all continents dispersed, which suggests that the break-up of Rodinia on the (paleo)north side should have begun at about that time. Our reconstruction places India and South China in a high-latitude zone for a long duration, despite no late Tonian glacial evidence on those two cratons. This may suggest a greenhouse climate during the Rodinia period. The pan-latitudinal distribution and the assembly kinematics of Rodinia resemble those aspects of Pangea, but they had different impacts on the Earth system during their assemblage and breakup. More paleomagnetic and geochronological studies from other continents are required not only to test the proposed TPW during this period but also constrain the position of each craton, which will lead to more sophisticated analyses of supercontinental paleogeography and its possible effects on the broader Earth system.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.epsl.2019.115880.

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